Multidisciplinary Reviews

BIBLIOMETRIC RESEARCH
Published Online: January 21, 2026
https://doi.org/10.31893/multirev.2026389

Artificial intelligence (AI) and flexible work arrangements: A bibliometric analysis with a systematic review

Şerife Durmaz[a] | **Aydın Çivilidağ**[b]

[a]Department of Labor Economics and Industrial Relations, Akdeniz University, Antalya, Türkiye.
[b]Department of Psychology, Akdeniz University, Antalya, Türkiye.

Abstract Artificial intelligence (AI) is a new technology developed through computer science. Today, artificial intelligence is starting to impact every aspect of human life, as well as business operations. In this research, artificial intelligence and flexible work arrangements were analysed using a bibliometric approach through a systematic review method. All article data related to "artificial intelligence" and "flexible work arrangements" in the Web of Science database were searched with a limitation to fields included in the social sciences. Although no year limitation was used in the research, the first article meeting the research criteria was published in 2014. The data for this research consisted of 60 articles conducted between 2014 and 2025. The selected dataset was analysed via Biblioshiny in the Bibliometrix R package. According to the results obtained, the main findings indicate a 23.28% annual increase in the number of articles related to concepts, a total of 320 author keywords, an average of 14.08 citations, and 25% of authors participating in international collaborations. The field with the highest number of articles in the dataset is management. The leading country is the USA. Sixty articles have been published in 55 different journals. Among institutions, the leading one is the Harvard Business School. In terms of thematic evolution, the keywords "remote work," "artificial intelligence," "coronavirus," and "digitalization" are the primary themes. In the historiographic analysis, three different paths were observed. Twenty-five percent of the publications in the dataset involve international collaboration, and an average of 14 citations per publication indicates international interest in the concept. This study provides a current and comprehensive overview of research on "artificial intelligence" and "flexible work arrangements," offering original and up-to-date information on the general trends and current standing of the field. This research contributes to the literature and provides valuable insights and a foundation for future studies.

Keywords: remote work, future of work, digital transformation, systematic review, labor relations

1. Introduction

Today, artificial intelligence and flexible working arrangements have led to significant changes in working life. Artificial intelligence and flexible working affect working life as a result of changes and developments in technology and social life. In the 21st century, AI-powered tools have become widespread in many sectors of working life, from marketing and sales to engineering and medicine (Kurtovic & Samardzija, 2023). Artificial intelligence has had significant impacts on corporate strategy, governance structures, employee relations, and organizational culture, as well as robotics and automation (Kraus et al., 2023). Artificial intelligence makes a significant contribution to the job performance of employees (Vaccaro et al., 2024). Research examining artificial intelligence in different cultures has shown that artificial intelligence has beneficial effects in terms of being helpful, productive, and supportive of employees. The same research revealed that employees have concerns such as fear of losing their jobs due to artificial intelligence (Kelley et al., 2021).

Like artificial intelligence, flexible working also offers various benefits to employees. Flexible working helps employees maintain a balance between their work and personal lives (Choudhary, 2016). Employees face not only professional challenges but also stress in their social lives and at home, as well as new and constantly changing challenges in today's dynamic work environment. Organizations that offer flexible working arrangements experience increased organizational support, trust, and productivity among their employees, which also helps reduce stress (Solanki, 2013). Flexible working arrangements have positive effects on employee performance, productivity, job satisfaction, job stress, work-family harmony, and organizational commitment (Çivilidağ & Durmaz, 2024). Research conducted via a social media application to understand attitudes toward remote working has also yielded partially positive results regarding remote working (Zhang et al., 2021). Although many studies mention both positive and negative outcomes, some positive outcomes include providing greater autonomy and flexibility among employees and reducing the likelihood of infection by reducing exposure to high-risk environments. Negative outcomes include increased anxiety, stress, and emotional exhaustion among employees, especially during the pandemic, leading to a

 Received: September 19, 2025 | Accepted: October 24, 2025

decline in overall health and well-being (Aleem et al., 2023). Another negative outcome is the inability to establish close, face-to-face relationships, compounded by misunderstandings caused by technological communication (Maguire, 2014). Although there are studies on the positive and negative aspects of artificial intelligence, remote working and artificial intelligence are clearly complementary rather than substitutive (Baldwin & Okubo, 2023). In recent years, both the COVID-19 pandemic and advances in technology have had significant impacts on working life. Flexible working has become an important alternative in terms of the sustainability of working life. Therefore, scientific research is needed to gain a more comprehensive understanding of the effects of flexible working and artificial intelligence on working life and employees. There is also a need for synthetic research on these two phenomena. The aim of this research is to examine studies that investigate flexible work and artificial intelligence via a systematic review method and bibliometric analysis. The absence of prior scholarly investigations in this specific area underscores the originality and potential contribution of the present study. Below, conceptual frameworks for flexible work arrangements and artificial intelligence are presented.

1.1. Literature framework

1.1.1. Artificial intelligence

The mathematician Alan Turing developed a code breaking machine called The Bombe for the British government, with the purpose of deciphering the Enigma Code used by the German army in the Second World War. The Bombe, which was a weight of approximately one ton, was generally considered the first electromechanical computer. Turing's 1950 publication, "Computing Machinery and Intelligence," articulated foundational concepts for developing intelligent machines and introduced a systematic approach—to evaluate their cognitive abilities. The word “Artificial Intelligence” (AI) was officially coined in the 1956 Dartmouth Summer Research Project on Artificial Intelligence in New Hampshire (Haenlein & Kaplan, 2019). Although artificial intelligence has developed primarily in the fields of mathematics and engineering, it has now begun to affect all human life, especially working life. In business life, the momentous rise of AI is a global revolution that is expected to persist in the future (Qamar et al., 2021). According to Boustani (2022), today's technology sector is undergoing a period of significant progress. Almost everything that used to be done by human labor has disappeared, and many jobs have been transferred to machines, software, various automated processes, and artificial intelligence. Artificial intelligence is nothing more than computers and machines developing intelligence similar to human intelligence. Artificial intelligence differs from the other four types of intelligence in terms of its historical development (Huang & Rust, 2018). This is shown in Figure 1. In addition, generative artificial intelligence has recently emerged as a separate type of artificial intelligence.

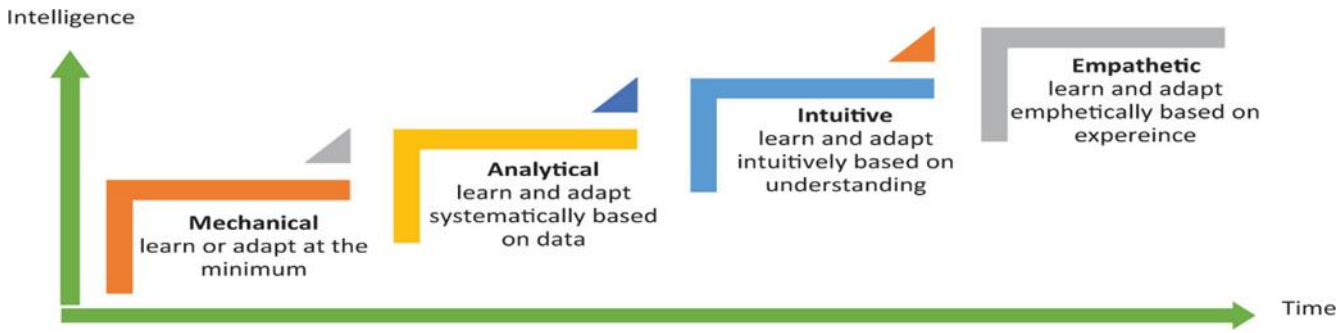

Figure 1 Four types of AI are cited from Huang & Rust (2018).

With respect to the transformation of artificial intelligence over time, Janiesch et al. (2021) defined machine learning algorithms (e.g., support vector machines, decision trees, and k-nearest neighbors), artificial neural networks (e.g., shallow approximate neighbors) and deep neural networks (e.g., convolutional neural networks and recurrent neural networks). While defining all three of these algorithms as machine learning algorithms, they grouped machine learning algorithms and artificial networks as shallow machine learning algorithms. They separated deep neural networks as deep learning. A Venn diagram of machine learning algorithm learning concepts and classes developed by Janiesch et al. (2021) was improved by Banh and Strobel (2023) by adding generative AI.This is shown in Figure 2.

With this technology, machines can perform simple and complex tasks that humans must perform regularly. Many factors may play a role in influencing employees' perceptions and experiences of AI in organizations. According to Leuhery (2024), technological competence, organizational support, and employee motivation mediate the relationship between technology adoption and training effectiveness. These factors can play important roles in shaping employee perceptions, skills, and engagement in technology-based training initiatives. AI has significantly impacted employees’ sentiments in the workplace, particularly against the backdrop of the recovery of work life from the pandemic. Although integrating AI into work life promises substantial benefits for the work life’s future, current and prospective employees may have varied reactions to the incorporation of AI and AI-powered technologies in their work environments (Gursoy & Cai, 2025). The Fourth Industrial Revolution (FIR), encompassing AI, robotics and machine learning, is predicted to have both positive and negative effects on

society, organizations, jobs, employee skills and competencies (Nankervis et al, 2021). Devices embedded with AI have become more common in various sectors. From an organizational perspective, these devices have several advantages. They can perform specific tasks faster and more precisely than can human beings (Singh et al., 2022). Generative artificial intelligence tools have the potential to enhance workplace performance and productivity, assess employee effectiveness, and foster a resilient, satisfied, and engaged workforce. Moreover, these tools can support the evaluation of human capital, skills, and career development; guide employee and productivity expectations related to flexibility and stability; and influence workforce turnover, retention, and loyalty (Kliestik et al., 2024). According to the findings of some studies, the relationships between the use of AI and employee performance, well-being, job satisfaction and job engagement are positive and significant (Almosawi et al., 2024; Chiong & Xie, 2024; Lestari et al., 2023; Sharif et al., 2025; Wijayati et al., 2022). In contrast, studies in the literature reveal employees' negative attitudes toward AI, for example, job insecurity, unemployment, job stress and burnout (Ali et al., 2024; Cheng & Jin, 2024; Dabbous et al., 2022; Hou & Fan, 2024; Kong et al., 2021; Sharma et al., 2024).

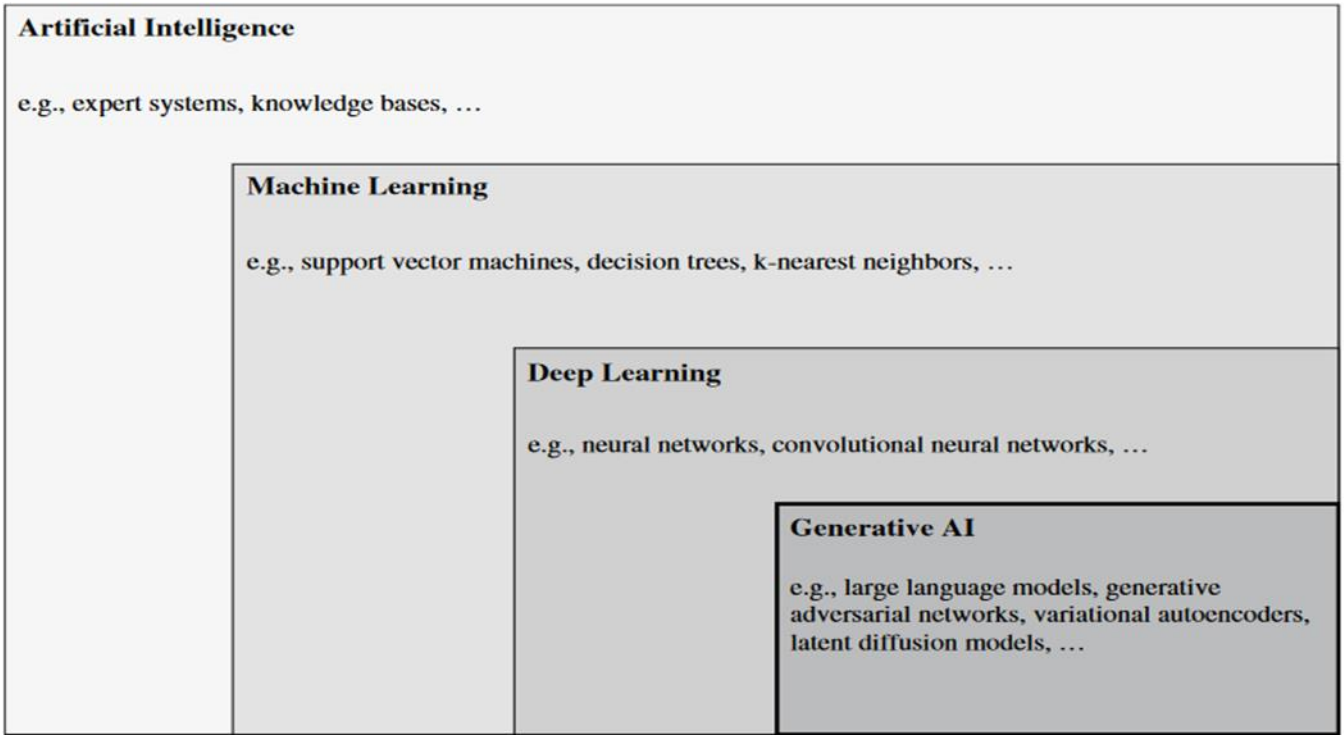

Figure 2 The development of generative artificial intelligence. *Source:* Banh & Strobel (2023).

The fear of machines taking over jobs, thereby making humans redundant, is not new. Since the Industrial Revolution, machines have been replacing blue-collar jobs. Considering recent developments in artificial intelligence (AI), is the end of white-collar jobs nearby, and how will this change the structure of the workforce? (Chelliah, 2017). This question appears to be an important source of concern among those working in the labor market. Both the uncertainty of the limits of artificial intelligence and how the labor market and employees react to this situation stand as problems that need to be researched. On the other hand, both the opportunities provided by technology and changes in social life affect work arrangements. Flexible work arrangements, which have become widespread due to the coronavirus disease 2019 (COVID-19) pandemic affecting the entire world, are now in greater demand by employees. Therefore, synthesizing studies conducted to better understand the impact of artificial intelligence and flexible work arrangements on employees and organizations is important.

1.1.2. Flexible work arrangements

Flexible working arrangements are designed to enable employees to change the amount, timing, or location of their work, allowing them to better balance their work and nonwork life demands. Examples include remote working, flexibility in working hours, and reduced working hours (De Menezes & Kelliher, 2017). Remote working is also referred to as telecommuting, teleworking, shared working, virtual working, flexible working, flexplace, and remote working, among other labels. While these various terms overlap, they generally encompass different conceptualizations of working from home (Allen et al., 2015). The concepts of flexible working and working autonomy refer to different situations. These two concepts are sometimes confused with each other. The main difference between flexible working and working time autonomy is that, unlike working time autonomy, flexible working still has some restrictions, such as having to stick to core hours (e.g., between 10 a.m. and 4 p.m.) and/or the number of hours employees can work in a day or week (e.g., 37 hours per week). Flexiplace, or tele- or home-based work, allows employees to work outside their normal workplace, such as from home (Chung & van der Lippe,

2020). Flexible working can provide employees with opportunities to devote time to themselves, their families, and the care and education of their children (Shagvaliyeva & Yazdanifard, 2014). Flexible working policies are designed to offer employees a degree of choice over how much, when, and where they work and to help them achieve a more satisfying work–life balance (Kelliher & Anderson, 2010). The accelerated development of information and communication technologies that transcend borders has led to more flexible forms of work organization, such as telecommuting and flexible working systems. Flexible work supports working outside the boundaries of an office at almost any time of day. Thus, flexible work enables employees to complete their work independently of time and location. This signifies a fundamental shift from traditional, external work arrangements to more internal, individual work arrangements within the boundaries between work and personal life (Mellner et al., 2014). Flexible working is both beneficial and problematic for employees. Its key benefits include allowing employees to spend more time with their families, preventing wasted time commuting to and from work, and enabling employees to control their working hours (Hayman, 2009; Subramaniam et al., 2015). Nonetheless, this approach is not uniformly applicable to all sectors. The capacity for remote work among employees varies considerably depending on the nature of their occupation and the characteristics of the sector in which they are employed (Adams-Prassl et al., 2022). One of the most significant potential negative consequences of working from home is the blurring of boundaries between work and family life. This can increase work–home interaction and lead to a poorer work–life balance (Russell et al., 2009; Kecklund et al., 2017; Chung & van der Lippe, 2020).

The widespread use of artificial intelligence technology in the workplace has become an important factor influencing flexible working arrangements. There is a need for further research to understand the impact of artificial intelligence on flexible operation. This research focuses on articles examining these two factors. Therefore, it aims to examine flexible work arrangements and AI with bibliometric analysis. In this context, the research question is presented below.

What are the bibliometric results of studies between AI and flexible working arrangements in the social sciences?

This research traces the historical development of the concepts of 'artificial intelligence' and 'flexible work arrangements' in the social sciences, offering a contemporary and comprehensive analysis of their evolution and current relevance. To this end, the following research questions have been identified to identify influential authors, journals, and studies in the field; map cooperation networks between countries; and identify key thematic areas.

1. In which fields of social sciences have studies been conducted on “artificial intelligence” and “flexible work arrangements”?
2. What has been the trend over time in articles and references to artificial intelligence and flexible work arrangements in the social sciences?
3. What are the leading articles and most cited authors in the social sciences in “artificial intelligence” and “flexible work arrangements” research?
4. Which journals publish the most articles on “artificial intelligence” and “flexible working arrangements” in the social sciences, and which journals receive the most citations?
5. Which country has published the most articles on “artificial intelligence” and “flexible work arrangements” in the social sciences, and which countries collaborate most frequently in these studies?
6. Which institutions publish articles on 'artificial intelligence' and 'flexible work arrangements' in the social sciences, in which journals, and which keywords are used?
7. What are the thematic findings of the keywords in articles on “artificial intelligence” and “flexible work arrangements” in the social sciences, and what are the future trends?

2. Methods

The concept of "bibliometrics," coined by Alan Pritchard, was first used in the Journal of Documentation in December 1969 (Broadus, 1987). In 1923, E. Wyndham Hulme proposed that the term "bibliometrics" be used in place of the term "statistical bibliography" used in the literature for the lectures he gave on bibliography at Cambridge University. Pritchard's definition characterized bibliometrics as utilizing mathematical and statistical methodologies to elucidate the mechanisms of scholarly communication and to trace the evolution and structure of academic disciplines (Lawani, 1981). Broadly conceptualized, bibliometrics refers to a suite of quantitative techniques applied to analyze patterns in scientific publishing and professional knowledge exchange, with the core objective of measuring various aspects of written communication (Diodato & Gellatly, 1994) .These techniques have been used to identify the convergence and division of scientific paradigms and disciplines and to identify the most productive journals and authors in different fields (Ikpaahindi, 1985). Bibliometrics encompasses all studies that use or discuss statistical analyses of data related to print communication (citation studies, abstracts of studies in journals, publication numbers, or individual elements within articles) (Pritchard & Wittig, 1981). Database coverage, data consistency and accuracy, data fields, search options, and analysis and use of metrics are fundamental elements of bibliometric analysis (Thompson & Walker, 2015). Bibliometric methods aim to estimate the extent to which a selected research article influences future studies (Cooper, 2015). The basic elements of bibliometric analysis include database scope, data consistency and accuracy, data fields, search options, and the analysis and use of metrics (Thompson & Walker, 2015). Bibliometric methods

estimate the extent to which a selected research article influences future research (Cooper, 2015). Bibliometrics is the application of quantitative methods to information studies (Brooks, 1996), and bibliometric analysis, as a quantitative measurement tool, is accepted as a literature review tool. Bibliometric analysis was preferred in this study to perform a general analysis of the quantitative findings. Different software packages can be used to analyse research questions, and in this research, the Bibliometrix software package is used to analyse the data. This package must be used in the R software platform. The bibliometrix is a comprehensive mapping analysis tool that supports three stages of the bibliometric analysis process: data import and conversion to R format, bibliometric analysis of a dataset, and matrix creation. Biblioshiny was selected for the analysis of the research data (Arruda et al., 2022).

2.1. Data collection and screening process

Historically, the first major change in the development of quantitative analysis in academic publishing was the creation of a citation database by the Institute for Scientific Information (ISI, now Thomson Scientific), which was established in 1961 (Thelwall, 2008). The Institute for Scientific Information (ISI) has been a leader in citation analysis since 1961, when it first published the Science Citation Index (McBurney & Novak, 2002). ISI's citation indices, now reorganized under Web of Science (WoS), have produced the only existing bibliographic databases that allow bibliometricians to compile large-scale bibliometric indicators (Archambault et al., 2009). Although databases such as Scopus have emerged in recent years, WoS has long been recognized as the most authoritative scientific and technical literature indexing tool providing data on the most important areas of science and technology research. Additionally, as a citation database, WoS provides sufficient search fields such as keywords, countries, institutions, authors, and references, which are particularly important for literature analysis in science-based analysis (Yao et al., 2014).

The data collection and screening process followed systematic methods to ensure reproducibility. The inclusion criteria were: (1) peer-reviewed journal articles, (2) publications within social sciences categories as defined by WoS, (3) documents containing both AI-related and flexible work-related keywords in their topic fields. Exclusion criteria encompassed conference papers, book chapters, editorials, and non-English articles without accessible English abstracts. Biblioshiny's default parameters were utilized for the science mapping analyses, with thematic mapping and network visualizations configured to optimally represent the core structure and relationships within the dataset.

Scientific articles related to the analysis were first searched without any limitations (all fields) via keywords such as “artificial intelligence” and “flexible work arrangements” from the core collection of WoS databases on June 23, 2025, as detailed in the flow chart in Figure 3 below. The identified keywords were subsequently searched via the “topic” filter. Topic searches include title, abstract, keyword plus, and author keywords. This approach ensured that the concepts did not appear in author names or journal titles, thereby yielding a more refined and purpose-specific dataset. The dataset was checked and verified by researchers before the scan continued. Since the aim of the research was to observe the development process of the concept, no year limitation was applied. Categories related to this field were selected because the research aimed to reveal the developmental process of both concepts in fields related to the social sciences. The categories are detailed in the flow chart. To ensure that the data appropriately address the research questions, the scope of the analysis was limited to documents classified as 'articles'. Other categories (conference papers, book chapters, etc.) were excluded. During the data validation process, one 'book chapter' that had been erroneously categorized under the 'article document type' was identified and subsequently removed from the dataset. As a result of the database search, 56 documents were initially classified as English, 3 as Russian, and 1 as Serbian. Upon manual verification, the article labelled Serbian was found to be written in English, indicating a metadata classification error, and was therefore corrected and retained in the dataset. For the documents categorized as Russian, the presence of search terms within the abstract and keywords was confirmed. Following a detailed content review, these documents were deemed relevant to the research scope and were consequently included in the final dataset. The screening and eligibility assessment were independently conducted by two reviewers. The consistency of the obtained dataset with the scope of the study was checked by the researchers one last time. The researchers agreed that every article in the final version of the dataset was within the scope of the study and that the dataset was created. A content screening process was conducted to ensure thematic alignment between the retrieved records and the scope of the study. Since all the articles demonstrated relevance to the identified research areas, no entries were excluded. Figure 3 provides a detailed flowchart for selecting articles for bibliometric analysis. Finally, a total of 60 articles were selected for the dataset. The dataset was downloaded in the form of “full records and references” (article title, author names and institutions and countries, journal names, article keywords, abstracts, publication years, and reference lists).

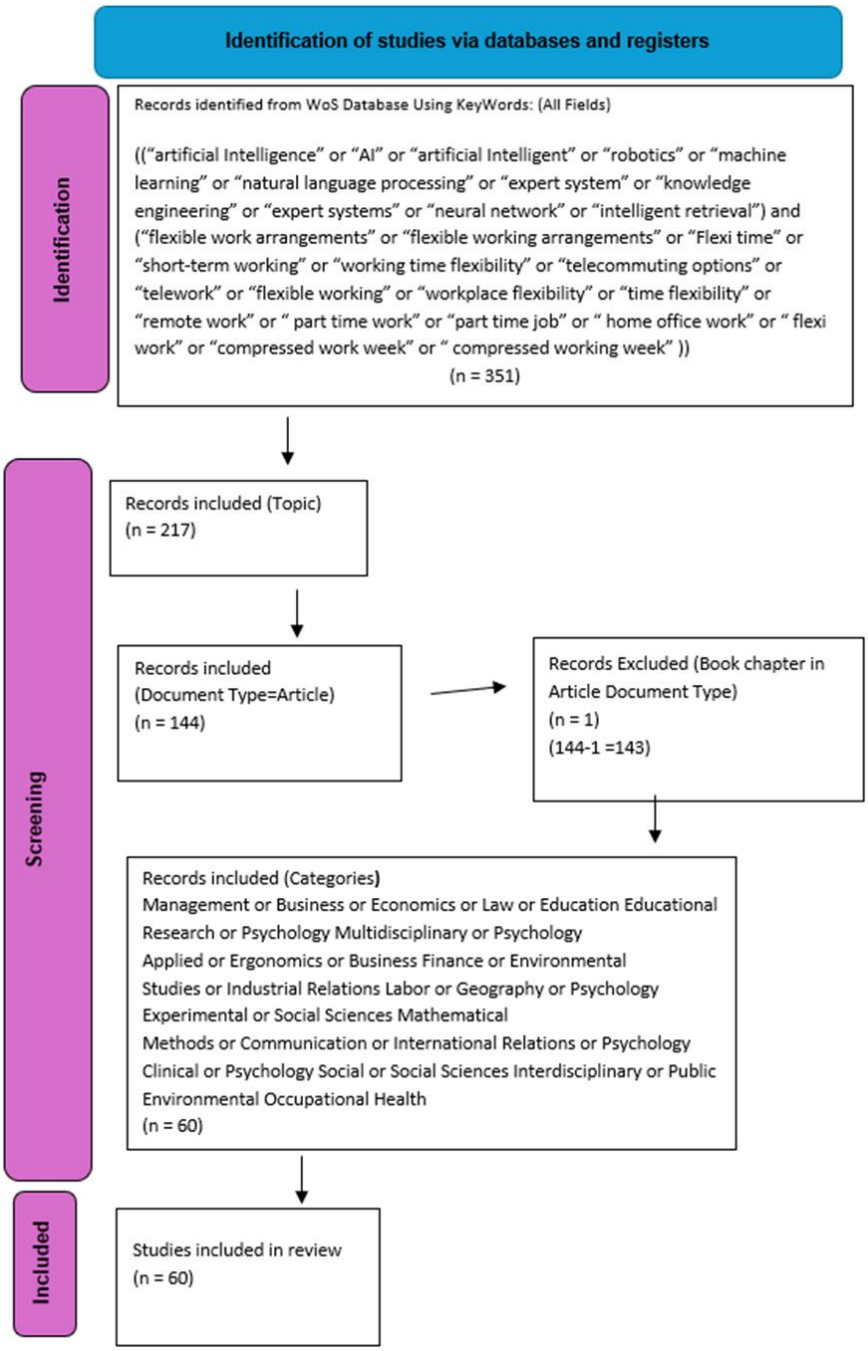

Figure 3 Flowchart of the investigations.

3. Results

Figure 4 presents a comprehensive analysis of the dataset before seeking answers to the research questions.

Figure 4 Main findings of the research. *Source:* Web of Science.

Despite the absence of a publication year filter during dataset construction, the analysis revealed that the concepts of 'artificial intelligence' and 'flexible work arrangements', which have become indispensable components of modern work life, have been the focus of a growing number of studies, especially in the last ten years. The earliest article included in the dataset was published in 2014. The dataset shows a 23.28% annual increase in the number of articles conducted in the social sciences on “artificial intelligence” and “flexible work arrangements.” This shows the growing interest in both concepts. The 60 articles in the dataset were obtained from 55 different journals. This makes us believe that many journals are open to publishing articles on these concepts. A total of 320 author keywords and 3,738 references were identified in all the documents. Among the 60 articles identified, there is an average of 14.08 citations per document. A total of 191 authors were identified in the dataset, 25% of whom were involved in international collaborations. In other words, one in four articles was coauthored internationally. There were only 14 single-authored articles. Each study has an average of 3.18 authors, meaning that each article was written by an average of three authors. Research on these two significant recent concepts is essential for assessing the scholarly influence, underlying framework, and prospective directions of work conducted in this field. Category field of the articles are given in Figure 5.

3.1. Category field of the articles

Figure 5 shows that the top 10 categories with the most articles in the study dataset are “Management”, with 16 articles, followed by “Business”, with 8 articles, and “Economics”, with 7 articles. Next are ‘Law’ with 6 articles, “Educational Research” with 5 articles, and “Psychology Multidisciplinary” with 5 articles. he fact that research has been conducted across nearly all social science disciplines indicates that these two concepts are critically examined from legal, economic, managerial, and psychological perspectives. Figure 6 shows the publication and citation processes of the articles included in the study over the years.Number of articles and average annual citations are given in Figure 6.

Figure 5 Category field with the most articles. *Source:* Web of Science.

3.2. Number of articles and average annual citations

The most cited article in the dataset is “Work-from-anywhere: The productivity effects of geographic flexibility” by Choudhury, Foroughi, and Larson, published in the Strategic Management Journal in April 2021, with a total of 242 citations. This article analyses the economic effects of work-from-anywhere in the context of geographic flexibility via microdata and machine learning.

Figure 6 shows that articles on “artificial intelligence” and “flexible work arrangements” in the categories identified in the Web of Science database in the field of social sciences began to be published in 2014. After 2014, when there was 1 article and 1 citation, the number of publications remained zero until 2020. The first publication within the dataset in 2014 was the article titled “Sociotechnical systems and interaction design--21st century relevance” by Maguire, M., published in the journal Applied Ergonomics in March 2014. The first citation for this study came from the article titled “Implementation of an integrated preoperative care pathway and regional electronic clinical portal for preoperative assessment” by Bouamrane and Mair,

published in November 2014 in the BMC Medical Informatics and Decision-Making journal. The first article published was in the field of ergonomics, which spans both engineering and social sciences, and the first citation to this study came from a journal that publishes articles on “the design, development, implementation, use, and evaluation of health information technologies and decision-making related to human health”. While there were no publications in the dataset between 2015 and 2019, Maguire, M.'s publication was cited once in 2015, five times in 2016, five times in 2017, seven times in 2018, and three times in 2019. By 2020, the number of articles had reached 2, and the number of citations had reached 6. One of these publications was a study titled “Usage and Impact of Artificial Intelligence on Accounting: Evidence from Malaysian Organizations” published by Lee and Tajudeen in the Asian Journal of Business and Accounting. After this article in the field of accounting, there was a significant jump in the number of articles published in 2021, with 9 articles and 35 citations. The articles published in 2021 covered a wide range of topics, including labor law, geographic flexibility, and the digital economy. In 2022, the number of articles and citations increased to 10 and 153, respectively. This trend continued in the following years, with 14 articles and 208 citations in 2023 and 15 articles and 250 citations in 2024. In the first half of 2025, there were 9 articles and 175 citations. The increase in the number of articles and citations over the past five years is likely to continue in the following years. The increase in interest in the field has also been reflected in academic productivity.

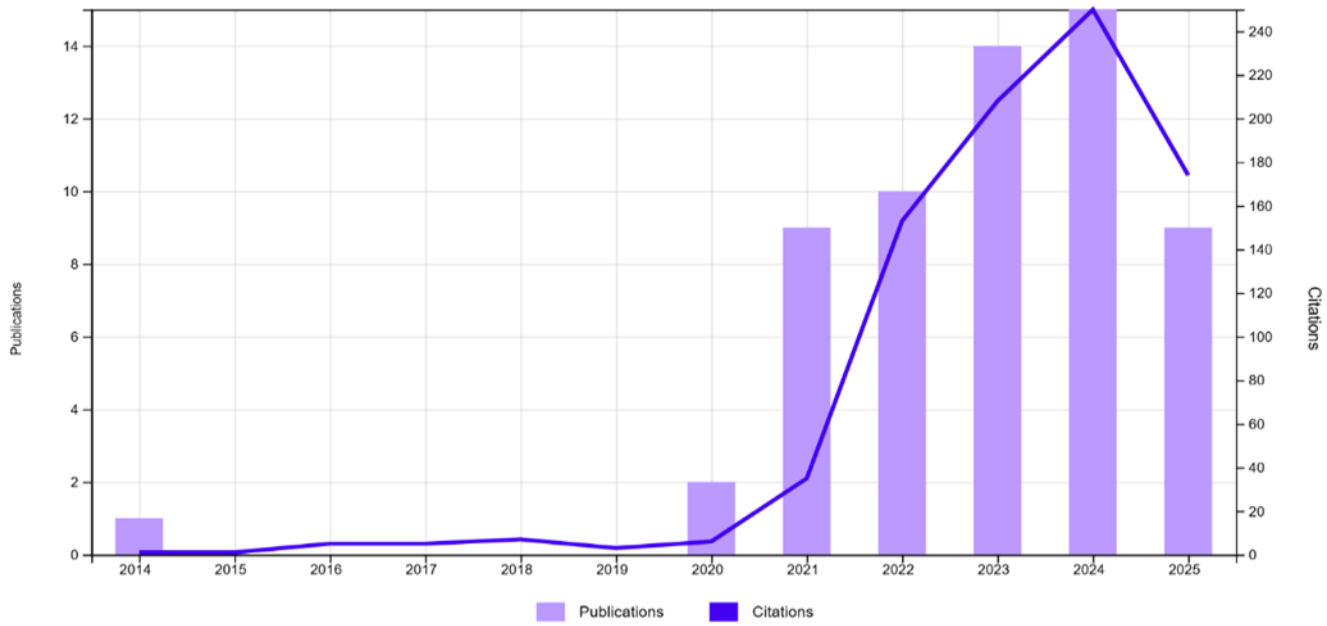

Figure 6 Number of articles and average annual citations.

3.3. Most local cited authors

Identifying the most cited authors in the fields of “artificial intelligence” and “flexible work arrangements” is important in indicating which authors and sources should be considered more carefully by new researchers. There is one article by each of the 191 authors in the dataset, and no author has more than one article. However, the number of citations to authors varies. When examining the local citation counts in the dataset in Figure 7, Bertello, Ferraris, Kraus, Marin, Yu, and Zhang each have two citations, whereas Adams-Prasl, Boneva, , Golin, and Rauh each have one citation. The other authors have no citations.

A detailed examination of these studies reveals that the article "The Future of Work: How Innovation and Digitalization Are Reshaping the Workplace" by Kraus, S., Ferraris, A., and Bertello, A., published in the Journal of Innovation & Knowledge between October and December 2023, received 47 global citations. While these citations do not originate from the same database or field, they indicate that these studies are cited by articles published in different fields or databases. In addition, this paper examines the "future of work" in the fields of business and management in four main research areas (remote and hybrid work, digital technologies and organizational applications, sustainable human resource management, and the future of organizations, sectors, and labor markets) and six thematic areas (organizational capabilities and culture, organizational performance, individual characteristics, knowledge management, the dark side of digitalization, and policy impacts). This also shows that there is a multisubject area related to concepts. The article titled “Exploring public sentiment on enforced remote work during COVID-19” by Zhang, C., Yu, M. C., and Marin, S., published in the Journal of Applied Psychology, was also published in 2021 and was the first article to stand out in the dataset. The article analysed the general public's attitudes toward and experiences with remote work via NLP techniques on Twitter keywords such as “telework,” “working from home,” “remote work,” and similar terms.

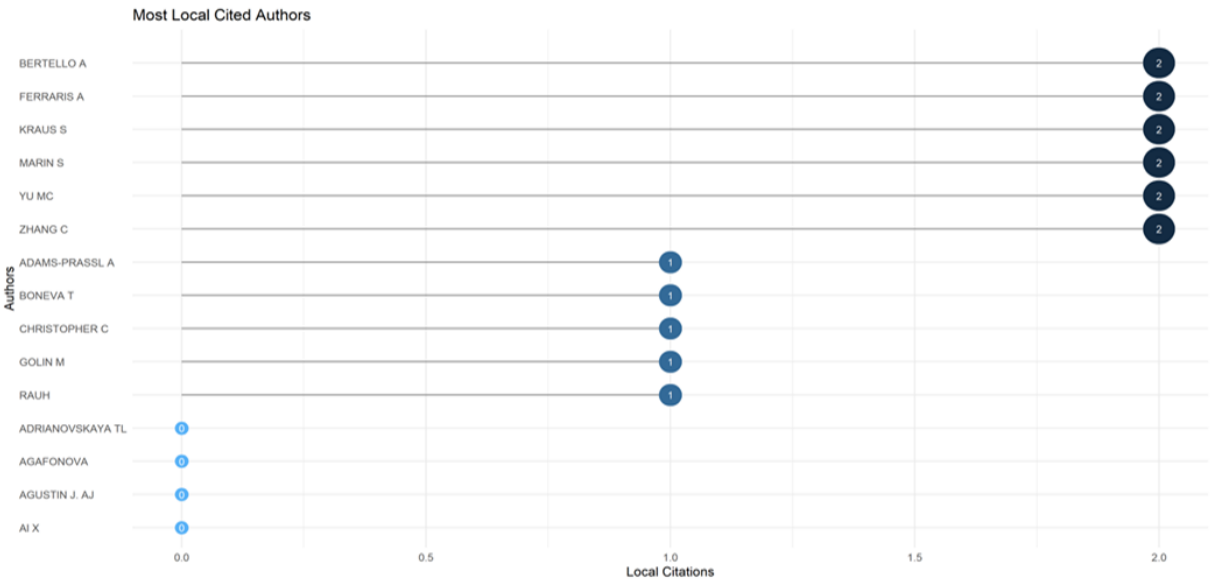

Figure 7 Most local cited authors. *Source:* Web of Science.

3.4. Most local cited sources

The journals most frequently cited in relation to the topics of "artificial intelligence" and "flexible work arrangements" are shown in Figure 8. However, it may be useful to examine the publication status of journals that have published articles on these concepts over the years to understand the citation process. As mentioned earlier, the first article in the dataset was published in 2014 in the journal "Applied Ergonomics." The journal is classified as Q2 in the SSCI and SSCI-E categories and covers the fields of engineering, industry, ergonomics, and applied psychology. There were no articles included in the dataset until 2020. In 2020, Lee and Tajudeen's article titled "Usage and Impact of Artificial Intelligence on Accounting: Evidence from Malaysian Organizations" was published in the Asian Journal of Business and Accounting. This journal is published in the ESCI field. In 2021, two articles were published in the journal "Sustainability," and one article each was published in "Economic Computation and Economic Cybernetics Studies and Research," "International Journal of Environmental Research and Public Health," "Journal of Applied Psychology," "Journal of Chinese Human Resources Management," "Journal of Legal Education," "Strategic Management Journal," "Vestnik Tomskogo Gosudarstvennogo Unıversiteta-Pravo-Tomsk State University," and "Journal of Law." In subsequent years, articles were published in other journals. The quarterly and index coverage of these journals varies. Sixty articles were published in 55 different journals. For this reason, it is difficult to say that any one journal stands out as a leader in terms of publication for this sample dataset. However, the number of citations to a publication is as important as the number of publications in terms of the visibility of the concept.

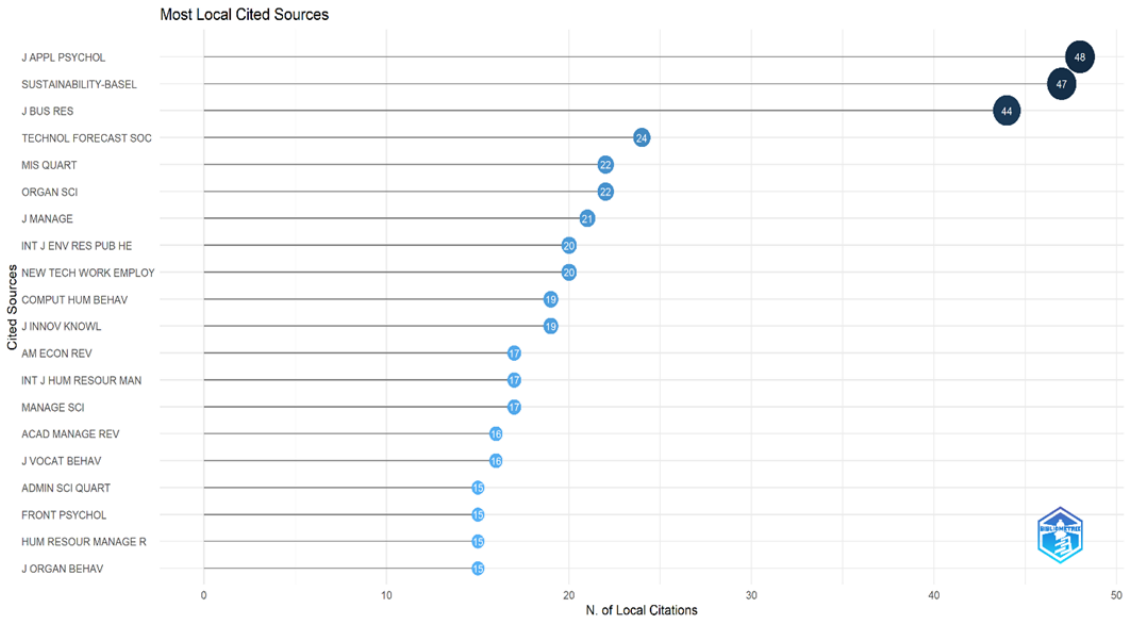

Figure 8 Most local cited sources. *Source:* Web of Science.

Although Applied Ergonomics was the first journal to publish articles on “artificial intelligence” and “flexible work arrangements,” the Journal of Applied Psychology became the leading journal in this field in 2021, with the publication of Zhang, Yu, and Marin's article “Exploring Public Sentiment on Enforced Remote Work During COVID-19.” With this publication, the journal received 48 local citations and 92 global citations, establishing itself as a leading journal. The next journal was Sustainability. This journal also published Lakner, Plasek, Kiss, Soós and Temesi's "Derailment or Turning Point? The Effects of the COVID-19 Pandemic on Sustainability-Related Thinking” and “Teleworking and Online Shopping: Socioeconomic Factors Affecting Their Impact on Transport Demand” by Lopez Soler, Christidis, and Vassallo, which were included in the dataset from 2021, has become one of the leading journals with 47 citations. The Journal of Business Research published one article in 2022 and one article in 2023 in the dataset and has nearly caught up with the citation counts of journals that published in previous years with 44 local citations. Other journals are ranked with fewer local citations. However, it would be beneficial for those who wish to study this concept to prioritize the journals listed in Figure 8. Country collaboration map is given in Figure 9.

3.5. Country collaboration map

At first glance, countries with dark colors stand out in terms of the number of articles produced. The number of articles decreases from dark to light colors. Accordingly, the country with the highest production is the USA with 31 articles, followed by Russia with 17 articles, China and Spain with 13 articles each, India with 11 articles, Hungary and Romania with 9 articles each, Switzerland with 8 articles, the UK with/article, Australia, Finland, Japan, and Poland with 6 articles each, Canada, Germany, and Malaysia with 4 articles each, the Czech Republic with 3 articles, Denmark, the Netherlands, New Zealand, and Italy, 2 articles each from France, Greece, Iran, Iraq, Lithuania, Mexico, Peru, Slovakia, South Africa, and Thailand, and finally 1 article each from Bolivia, Jordan, and Oman. From this, it is clear that the most productive country is the USA.

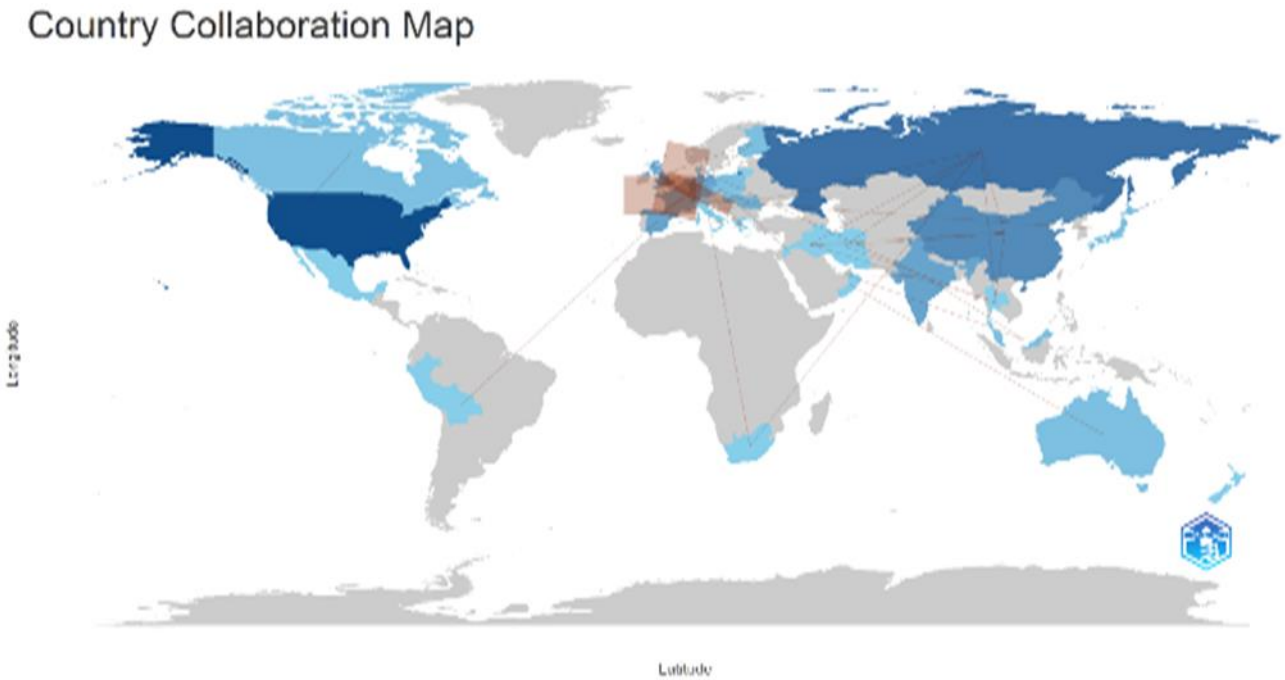

Figure 9 Country collaboration map. *Source:* Web of Science.

There are eight clusters in international cooperation. The first cluster includes Russia, China, Italy, Iran, Iraq, South Africa, and Thailand; the second cluster includes Switzerland, the United Kingdom, Finland, Japan, Germany, the Netherlands, France, and Greece; the third cluster includes India, Malaysia, Denmark, and the United Arab Emirates; the fourth cluster includes Romania, the Czech Republic, and Slovakia; the fifth cluster includes Spain and Bolivia; the sixth cluster includes Australia and Oman; the seventh cluster includes Hungary and Jordan; and finally, the eighth cluster includes the USA and Canada.

When considering the international cooperation network, the network only shows countries that have at least one cooperation (i.e., one connection). The gray-colored countries indicate isolated countries, i.e., countries without connections. Countries not shown in the table have a frequency of zero and are not visible on the map. Visually, edge size (ES=50) provides a fixed line thickness for all connections, whereas line thickness appears independently of frequency. This makes it difficult to distinguish between intense and low levels of cooperation. The dark or light coloring of countries is based on the total frequency of country connections. The intercountry collaboration network shows the intensity (frequency) of scientific collaboration between two countries. A high value indicates strong collaboration between those countries in terms of joint publications and projects.

The highest level of cooperation in the dataset is between Switzerland and Japan (freq=138), India and Malaysia (freq=109), and Russia and China (freq=103). There appears to be continuous and active cooperation between these countries. However, negative values between some countries (Germany–France: –2.76, US–Canada: –98) may be interpreted as data quality issues, bidirectional thresholds, or calculation method deviations, resulting in meaningless values. Overall, the figure indicates collaboration at both the regional and global levels. For example, collaborations between neighboring countries such as Romania–Slovakia (freq=19) reflect the influence of geographical proximity, whereas Russia, which shares the same frequency value (freq=101) with Thailand, Iran, and Iraq, highlights intercountry academic coordination.

The countries with the highest level of collaboration are Switzerland and Japan (freq=138). These countries are leaders in technology and engineering, which enables them to collaborate in these fields. The second most intensive collaboration is between India and Malaysia (freq=109), representing the scientific proximity between South Asia and Southeast Asia. The collaboration between Russia and China (freq=103) shows that geopolitical rapprochement in the fields of politics and technology has also translated into scientific collaboration. The cooperation between China and Thailand (freq=101) may also be a result of China's scientific diplomacy with Southeast Asian countries, given its position as a central country. The high level of cooperation between Iran and Thailand (freq=101) is important in terms of knowledge transfer between developing countries. Thailand's openness to multilateral cooperation is also reflected in its cooperation with Iraq (freq=101) and Russia (freq=101). Other collaborations are at a medium or low level and involve different country partnerships [Australia and Oman (freq=56), India and the United Arab Emirates (freq=54), Russia and South Africa (freq=25)].

The collaboration scores used in the Collaboration World Map analysis are normalized or based on statistical models. Therefore, in some country pairs, publication density may fall below the expected level, resulting in negative scores (Spain and Bolivia (freq= -64.69), USA and Canada (freq= -98.31)).

3.6. Three-field graph analysis

Different analyses and mappings are combined to identify and visualize subareas related to the defined concepts. This research enables the quantitative measurement and visualization of the evolution of the field, thematic areas are used to show conceptual evolution, and different diagrams are used to categorize and graphically display the identified themes (Cobo et al., 2011). Sankey diagrams are flow diagrams that were originally developed in the field of engineering to show energy flows. While providing quantitative information about flows, relationships, and transformations, these diagrams feature directed, weighted graphs with weight functions that ensure flow conservation. Here, the sum of the incoming weights for each node is equal to the outgoing weight (Riehmann et al., 2005). This diagram, which began to be used in many different fields over time, was later also used for bibliometric analyses. Three-field plot analysis is particularly useful for visualizing the relationships among the main elements of three fields. In this research, it was used to visualize which institutions preferred which keywords and in which journals they published regarding the concepts identified. Figure 10 below visualizes the most frequently used keywords in the dataset and the connections between institutions and publication sources. For clarity, the number of items for each variable in Figure 10 is set to 15. Institutions are shown on the left, author keywords in the middle, and sources on the right.

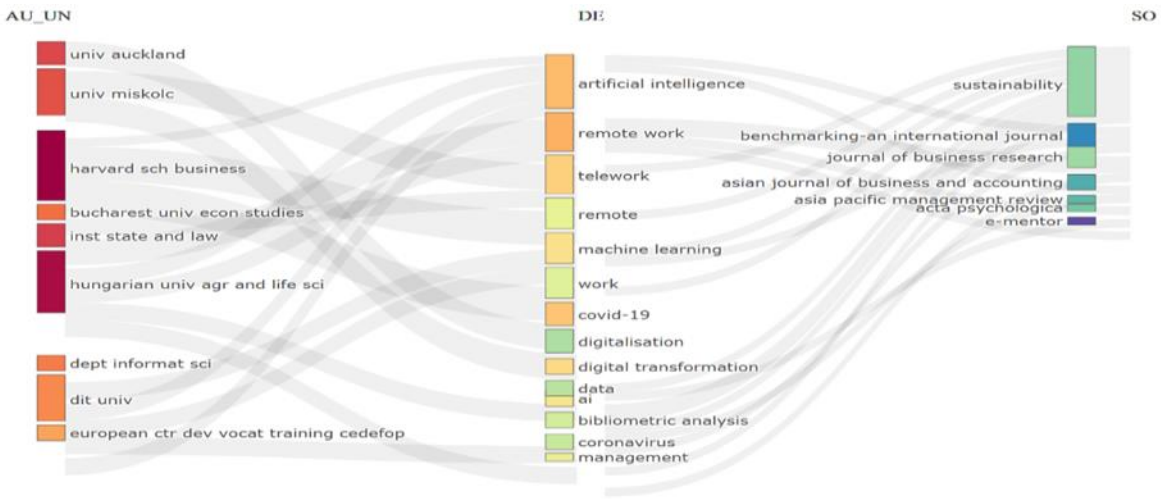

Figure 10 Three-field plot. *Source:* Web of Science.

Figure 10 shows that the journal that most frequently uses the 8 author keywords (telework, machine learning, remote, work, digital transformation, coronavirus, data, AI) is “sustainability”. Benchmarking, an international journal with "artificial intelligence", "machine learning" and "bibliometric analysis" keywords, the Journal of Business Research with "remote work" and "management" keywords, Asian journal of business and accounting with "artificial intelligence" and "AI" keywords, Asia

Pacific Management Review and e-mentor journals with "remote work" keywords and finally Acta Psychologica with "digital transformation" keywords were included. Among the institutions, the leading institution is the Harvard School of Business. While this institution stands out with the keywords "remote", "work" and "artificial intelligence", the Hungarian University of Agriculture and Life Sciences stands out with the keywords "artificial intelligence", "remote work", "coronavirus" and "data", and the University of Miskolc stands out with the keywords "telework" and "digitalization".

3.7. Thematic maps

Another technique used to discover the conceptual structure of variables (keywords, topics, abstracts, etc.) in a dataset is thematic analysis. Thematic analysis, which is used to reveal trends, prominent topics, and their temporal evolution in a given field, is evaluated in two basic dimensions. Centrality shows the importance of a topic in the field by measuring the strength of the connection between the defined theme and other themes. Density indicates the internal development and consistency of a theme, i.e., its level of maturity. The interplay between these two dimensions yields a classification into four distinct theme categories: motor themes, niche themes, emerging or declining themes, and basic themes. Motor themes are powerful themes that drive the development of the field, with high centrality and density, and are located in the upper right quadrant of the field. Niche themes, on the other hand, represent a situation with high density and low centrality in the upper left quadrant. These include themes that are technically advanced but have yet to make a significant contribution to the main field. Emerging or declining themes, located in the lower left quadrant, have low density and centrality, meaning that they are either just beginning to emerge or are losing their influence. Basic themes are shown in the lower right corner and have high centrality and low density, meaning that although they form the basis of the field, they are not yet sufficiently mature (Callon et al., 1983, 1991; Aria & Cuccurullo, 2017). Figure 11 below shows a thematic map prepared according to author keywords.

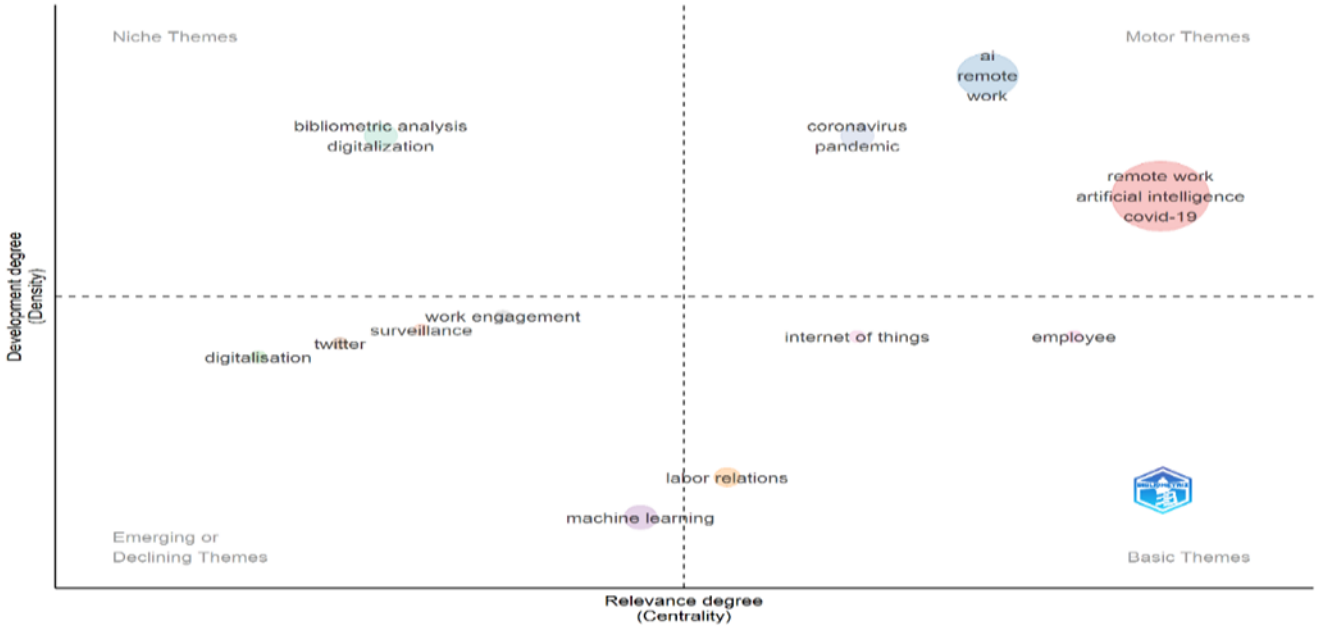

Figure 11 Thematic map. *Source:* Web of Science.

There are 12 clusters on the thematic map. The label names are given according to the highest occurrence. The cluster names are remote work, AI, digitalization, machine learning, labor relations, Twitter, employee, work engagement, bibliometric analysis, surveillance, coronavirus, and Internet of Things. The remote work cluster consists of the words remote work, artificial intelligence, COVID-19, telework, digital transformation, management, future of work, human resources, hybrid work, and work from home. The AI cluster consists of the words AI, remote, work, data, and teamwork. The coronavirus cluster consists of the words coronavirus and pandemic. The digitalization, machine learning, labor relations, employee, Twitter and work engagement clusters consist of single words that bear their own names.

Remote work, artificial intelligence, coronavirus, and digitalization are the main themes. These keywords can be studied in the future and contribute to the field. Digitalization and bibliometric analysis are niche themes, and their contribution to the field can be increased through further study. The Internet of Things, labor relations, and employees are basic themes. Because these concepts are more general, they are considered basic themes. This is because concepts such as labor relations and the Internet of Things contain many more subconcepts, and authors often express these concepts with more microconcepts (AI, remote work). Therefore, studies on more general concepts can be carried out to bring them to the forefront within these main themes. Surveillance, work engagement, and machine learning are emerging or declining themes. Among these keywords,

surveillance and work engagement are particularly prominent in terms of both density and centrality. Interest in concepts falling under this theme is expected to decrease or increase over time.

Thematic evolution can simplify and make the transformational processes of themes more visible. Figure 12 shows the thematic evaluation.

As shown in Figure 12, several studies have been conducted since 2024. The themes between 2014 and 2023 have changed as of 2024. The impact of the pandemic is particularly evident here. There is a strong focus on concepts such as remote work, artificial intelligence, and COVID-19. This situation is expected to lead to a conceptual transformation in the coming periods.

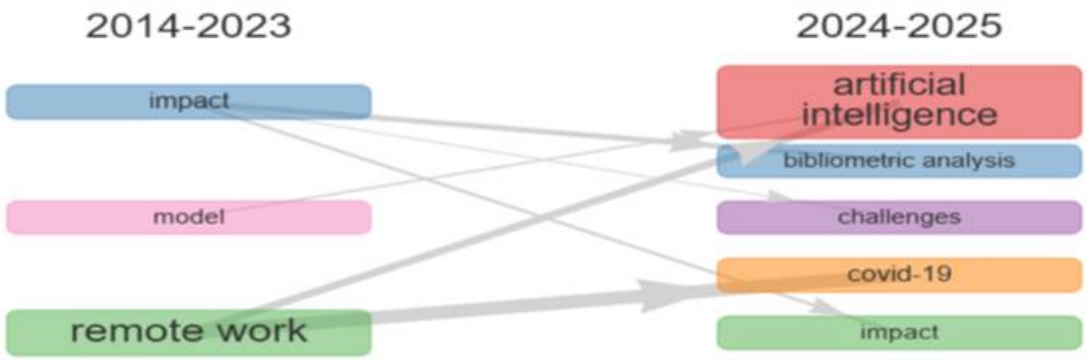

Figure 12 Thematic evolution. *Source:* Web of Science.

3.8. Historiographic analysis

Historiographic networks are used to visualize the developmental process of a dataset and the paths of knowledge transfer. While the details of individual datasets have been explained in detail above, this analysis is useful for visualizing the overall transformation. The historiograph shows not only the interaction of studies within a given field over time but also their sequential relationships. This network, constructed through citation relationships, shows which key studies (nodes) influenced which studies or were followed by which publications. In other words, each node represents an article, and each arrow represents the number of times this publication cited another study. These graphs, used to understand a chronological process within the field, such as which publications influenced subsequent studies and how the chosen field has evolved, display older studies on the left and more recent studies on the right. The arrows (citation relationships) show which studies were influenced by or cited by each other, allowing us to visualize the mainstream transformation. In the graph, the X-axis represents time, whereas the Y-axis represents the local citation score (LCS). The placement of a publication depends on the LCS it receives from other publications and can be on the right or left of the graph (Garfield, 2001, 2004; Garfield et al., 2003; Bar-Ilan, 2008). The global citation score (GCS) is also given in the table. Histographic map is given in Figure 13.

Figure 13 Histographic map. *Source:* Web of Science.

Figure 13 shows three different information flows (paths), with different colors indicating citation clusters. The timelines are again determined according to the publication year, either from top to bottom or bottom to top. Zhang et al. (2021) is at the center of the red cluster, and this paper focuses on the core of the cluster. The subsequent studies by Saura et al. (2023) and Aleem et al. (2023) were directly influenced by this paper, meaning that they cited it. Kraus et al. (2023) is at the center of the green cluster and influenced the papers by Kliestik et al. (2024) and Bhardwaj et al. (2025). Adams-Prassl et al. (2022) is at the center of the blue cluster, and this work was also cited by Baldwin and Okubo (2023), demonstrating a sequential relationship. The lack of cross-cluster citations indicates that these three structures have grown independently of each other as separate themes. Therefore, it can be said that growth has occurred in a multicentered structure.

The historiographic map shown in Figure 13 is given in more detail in Table 1 below.

Table 1 Historiograph network.

Paper	Title	Author_Keywords	Year	LCS	GCS	cluster
Zhang et al., 2021, J Appl Psychol. doi 10.1037/APL0000933	Exploring Public Sentiment On Enforced Remote Work During Covid-19	remote work; twitter; web scraping; covid-19	2021	2	92	1
Saura et al., 2023, J Bus Res. doi 10.1016/J.JBUSRES.2021.12.052	Exploring The Challenges Of Remote Work On Twitter Users' Sentiments: From Digital Technology Development To A Post-Pandemic Era	remote working; sentiment analysis; computer-aided text analysis; topic; modeling; twitter; ugc	2023	0	109	1
Aleem et al., 2023, J Bus Res. doi 10.1016/J.JBUSRES.2022.113303	Remote Work And The Covid-19 Pandemic: An Artificial Intelligence-Based Topic Modeling And A Future Agenda	remote work; work from home; employee; organization; human resource; management; literature review	2023	0	29	1
Adams-Prassl et al., 2022, Labor Econ. doi 10.1016/J.LABECO.2021.102083	Work That Can Be Done From Home: Evidence On Variation Within And Across Occupations And Industries	working from home; occupations; industry; coronavirus; covid-19; telework	2022	1	63	2
Baldwin and Okubo, 2023, World Econ. doi 10.1111/TWEC.13496	Are software automation and teleworker substitutes? Preliminary evidence from japan	ai; complement; covid-19; globotics quadrant diagram; labor; ri; substitute; telemigration; work from home	2023	0	5	2
Kraus et al., 2023, J innov knowl. doi 10.1016/J.JIK.2023.100438	The Future Of Work: How Innovation And Digitalization Re-Shape The Workplace	future of work; ai integration; digital transformation; disruptive; technology; remote work; covid-19; knowledge work; human resource; management; organizational culture; skill development; policy; implications	2023	2	45	3
Kliestik et al., 2024, Oecon Copernic. doi 10.24136/OC.3109	Enterprise Generative Artificial Intelligence Technologies, internet Of Things And Blockchain-Based Fintech Management, And Digital Twin Industrial Metaverse In The Cognitive Algorithmic Economy	enterprise generative artificial intelligence; internet of thing; blockchain; fintech; digital twin industrial metaverse; cognitive; algorithmic economy	2024	0	5	3
Bhardwaj et al., 2025, Int J Organ Anal. doi 10.1108/IJOA-10-2024-4895	Navigating The Digital Evolution Of HRM: An Integrative Review	human resource management; digitalization; strategic HR; integrative; review; bibliometric analysis	2025	0	0	3

Articles belonging to the red cluster (cluster 1) examine remote work through social media data (Twitter) and text mining (sentiment analysis). Zhang et al. (2021) conducted a leading article analysing remote work during the pandemic via social media data, whereas other authors, Saura et al. (2023) and Aleem et al. (2023), continue this methodology. In the blue cluster (cluster 2), Adams-Prassl et al. (2022) conducts an empirical study on the workforce, whereas Baldwin presents a theoretical approach that addresses remote work in the context of global labor dynamics. In other words, this cluster has a separate structure that addresses the macroeconomic effects of remote work and the effects of artificial intelligence and labor markets. The green cluster (cluster 3) includes articles published via different methods, such as systematic review and bibliometric analysis. The articles in this cluster include recent studies that address the impact of artificial intelligence and digital transformation on human resources. The clusters provide a clearer understanding of the transformation process and conceptual changes in the field. Additionally, as previously mentioned, the citation analysis was evaluated via LCSs. The table

shows that Saura et al. (2023), which is not an LCS, has the highest number of GCSs. Therefore, conducting citation studies with broader databases could lead to different results.

However, as mentioned above, few databases provide the dataset needed to answer other research questions within the scope of this study. For this reason, the WoS database was chosen for its reliability, academic acceptance, and dataset.

4. Discussion

By exploring the trajectory of 'artificial intelligence' and 'flexible work arrangements' within the social sciences, this study provides an updated and integrative perspective on how these concepts have evolved and been theorized over time. The research findings reveal academic interest in two concepts that have come to the forefront in recent years: “artificial intelligence” and “flexible work arrangements.” Although there is no year limitation in the research dataset, interest in these concepts has increased significantly, especially in the last ten years. The first article in the dataset was published in 2014, and there has been a 23.28% annual increase in the number of articles. The large number of journals in the dataset also indicates that journals are open to publishing articles on these concepts. The fact that 25% of the articles in the dataset involve international collaboration and that there are an average of 14 citations per article further demonstrates international interest in the concept.

Although a limitation was imposed on the social sciences field when creating the dataset, the social sciences field itself contains many subfields. This variety is also reflected in the dataset. Although artificial intelligence was initially the subject of fields such as engineering, as seen from the dataset, it has become a research topic in almost every field of the social sciences. While management is the field with the most publications in the dataset, research has also been conducted in fields such as economics, law, education, and psychology.

Although there is no time restriction on the dataset, the articles that comprise it were published between 2014 and 2025. The number of articles related to these concepts has increased since 2020 and has accelerated considerably since 2021. Although this concept has existed for a long time, the COVID-19 pandemic that emerged in 2019 can be cited as a reason for the increased prominence of flexible work arrangements. Similarly, the increased prominence of deep learning in artificial intelligence since 2020 may have contributed to the increased prominence of research on these two concepts.

The sharp increase in publications from 2020 onwards, as revealed by the bibliometric data, can be attributed to two major global catalysts. First, the COVID-19 pandemic acted as a massive, unplanned experiment in remote work, forcing organizations worldwide to adopt flexible work arrangements overnight (Kniffin et al., 2021). This created an urgent practical need and a rich new context for research at the intersection of work and technology. Second, concurrent breakthroughs in generative AI and deep learning, particularly the rise of large language models, moved AI from a backend tool to a pervasive force with immediate implications for knowledge work and employment (Dwivedi et al., 2023). The convergence of these two disruptive forces—a global shift in where we work and an acceleration in how we work—explains the observed surge in scholarly attention. This finding aligns with the concept of the "Future of Work" as a research domain that is highly sensitive to exogenous technological and social shocks (Kraus et al., 2023).

When the articles in the dataset are examined, they focus more on the effects of technological developments in artificial intelligence on labor markets and organizational structures. Although the articles are published in different journals, they focus more on certain basic areas. For example, both human resource management and labor economics journals related to working life have come to the forefront. Journals publishing studies on the effects of artificial intelligence and the workforce on the digital dimension, as well as the impact of digitalization technology on the changing workplace, also stand out. In addition to a wide variety of journals, the articles in the dataset cover different areas. Articles on strategic management and business innovation, as well as ergonomics, remote work, psychology, and technology, are included in the dataset. When examining the overall picture of the articles comprising the dataset, it is possible to say that they focus primarily on the effects of digital transformation and artificial intelligence on organizational dynamics, strategic management, and workforce psychology and that technology and workforce adaptation are increasingly becoming areas of focus.

The number of citations for authors who have received the most citations in the fields of “artificial intelligence” and “flexible work arrangements” is quite low. This may be because the dataset consists of articles published in the last ten years. Because an average document age of two is considered very new for the social sciences, the citation process can take longer than in the natural sciences. Furthermore, bibliometrics is a field applied to disciplines based on data obtained from databases where literature and citations are appropriately organized, which puts the social sciences at a disadvantage. This is because the natural sciences are better represented in such databases, whereas the literature on the technical sciences, social sciences, and humanities is only partially included in these databases. Bibliometrics can provide only limited results for these disciplines. Therefore, although the data from the WoS database, which is considered the most reliable database, were used in this research, some inadequacies may have emerged in reflecting the field of the dataset. Additionally, the average time it takes for a paper to receive citations ranges from 2--4 years. Since citations accumulate over time, bibliometric-based research evaluations cannot make many comments about new publications. Because it is standard practice in bibliometric analysis to allow at least three years to reliably measure the impact of a publication (Bornmann & Leydesdorf, 2014), sufficient time may not yet have elapsed for citation accumulation. Furthermore, these results are local citation data, and the number of citations

is limited because of citations from other publications in the dataset. Global citation data may differ. These studies may also have been cited from databases other than WoS. Furthermore, the existence of many subject areas that include these two concepts may also contribute to the low number of citations. In summary, the newness of the publications, the wide variety of subjects that include these concepts, and the small size of the dataset may have contributed to this. It is expected that the number of citations will increase in the following periods.

Sixty publications were published in 55 different journals. Therefore, it is difficult to say that this journal is a pioneer for this sample dataset. However, the number of citations to a publication is as important as the number of publications in terms of the visibility of the concept.

In 2021, the Journal of Applied Psychology, which published Zhang, Yu, and Marin's article “Exploring Public Sentiment on Enforced Remote Work During COVID-19,” became the leading journal with the most local citations. The main reason for this may be that the first publications received more citations. As mentioned earlier, publication frequency began to increase in 2021, and since the articles published in 2021 were leading articles on conceptual coherence, it was expected that they would stand out in terms of local citations. However, the fact that articles published in subsequent years also achieved similar citation scores indicates that the leading journals and articles may change in the future.

Social developments and circumstances influence the emergence and development of a concept. Unsurprisingly, certain concepts emerge in regions that have undergone specific historical processes. For this reason, examining the country that publishes the most in a given field and the state of cooperation between countries is helpful in understanding conceptual development. This is because the central region is influential in the spread of academic trends and studies. Within the scope of the research, which countries place greater importance on conducting research, which countries’ citizens publish more articles in these fields, and which countries engage in more intensive cooperation were examined. The United States is the leading country in this regard. The highest levels of cooperation were observed between Switzerland and Japan, India and Malaysia, and Russia and China. The highest-frequency relationships are centered around the Europe–Asia and Asia–Asia pairs. In this case, it is consistent with the “hub-and-spoke” structure, where central countries collaborate more with other central countries, while peripheral countries remain further removed from these connections.

The finding that the United States is the most prolific country in this domain is consistent with its established role as a central hub for research in both management and technology-driven fields. This leadership can be attributed to several factors, including substantial investment in research and development, a high concentration of leading technology firms and academic institutions, and the early and widespread adoption of flexible work arrangements, particularly during the pandemic (Kniffin et al., 2021). However, the emergence of strong collaborative clusters in Europe-Asia and Asia-Asia pairs, such as Switzerland-Japan and Russia-China, signals a shifting and increasingly polycentric global research landscape. This suggests that while the U.S. currently leads in volume, the intellectual influence and future trajectory of the field are becoming more geographically distributed.

The collaboration scores used in the Collaboration World Map analysis are normalized or based on statistical models. Despite publishing extensively in this field, collaboration networks with other countries may be statistically low. This can sometimes lead to negative statistical data. For example, countries such as Spain, Bolivia, the USA, and Canada are positioned as isolated or marginal nodes in the network. In other words, these are countries that show limited interaction in terms of scientific collaboration.

The fact that the journals that stand out in the three field plot analyses are journals that publish more in fields such as business and psychology is a good example of the convergence of the mechanical world and the social world. The fact that the Harvard School of Business is a leading university in scientific progress demonstrates the importance of the relationship between these concepts. The existence of institutions from different countries in this field shows that these concepts are taken into consideration by many institutions.

The themes that are expected to be studied in the thematic analysis in the coming days and contribute to the field are motor themes. Remote work, artificial intelligence, coronavirus, and digitalization are key words that appear in motor themes. Digitalization and bibliometric analysis are niche themes, and further research in these areas could increase contributions to the field. The Internet of Things, labor relations, and employee words are considered basic themes because they are more general concepts. Concepts such as labor relations and the Internet of Things contain many more subconcepts within themselves, and authors often express these concepts using more micro concepts (AI, remote work). For this reason, perhaps studies focusing on more general concepts could bring these overarching themes to the forefront. Surveillance, work engagement, and machine learning are emerging or declining themes. Among these concepts, surveillance and work engagement stand out in terms of both density and centrality. Interest in the concepts within this theme is expected to decrease or increase over time. As authors, our view is that these concepts will become even more prominent. The digital world, which is becoming increasingly prevalent in our lives, is expected to continue to grow in importance in the workplace. Flexible work arrangements, which have rapidly increased, especially in the postpandemic period, are now affecting a significant part of the workplace. In the context of the impact of these concepts, surveillance and work engagement will be discussed in particular. From 2024 onwards, there was a break in the studies, and the themes changed as of 2024. The impact of the pandemic is particularly evident here. There is a strong focus on the concepts of remote work, artificial intelligence, and COVID-19. In this

case, it is expected that a conceptual transformation will take place in the coming periods. The identification of 'remote work,' 'artificial intelligence,' 'coronavirus,' and 'digitalization' as motor themes is particularly significant. This finding aligns with broader scholarly discourse that identifies the pandemic as a critical juncture that accelerated the adoption of both remote work models and AI-driven technologies, effectively merging these research streams (Kniffin et al., 2021; Dwivedi et al., 2023). While a prior bibliometric review by Subramony et al. (2023) identified "automation," "digital technologies," and "gig work" as distinct research clusters, our analysis reveals that these themes have now coalesced with 'remote work' into a unified, powerful research front. This observation corroborates the findings of Singh et al. (2022), who in their systematic review, highlighted that the COVID-19 pandemic acted as a powerful accelerator, fundamentally reshaping and merging pre-existing trends in the future of work literature. This collective evidence from multiple bibliometric studies suggests a definitive paradigm shift, moving from examining technological and organizational concepts in isolation to studying their synergistic effects on the future of work.

The histiographic network visualizes the interdisciplinary thematic development and temporal transformation of selected concepts. Three distinct clusters emerge in the histiographic visual. These clusters reveal the intellectual reflections of a specific scientific community that addresses different methods and topics. Therefore, each cluster combines different conceptual approaches and methodological preferences. The central study of the first cluster, Zhang et al. (2021) has a decisive impact in terms of digital transformation/artificial intelligence applications. Kraus et al. (2023) central cluster influences the literature by focusing on economic or managerial impacts. The other cluster, centered around Adams-Prassl et al. (2022), addresses the effects of remote work on the labor market, industrial transformation, and global labor mobility. This study highlights the transformative impact of the pandemic and emphasizes the complementary and substitutive effects of digitalization on the labor force. This cluster shows that remote work must be examined not only in terms of its individual and organizational structure but also in terms of socioeconomic systems. The first cluster addresses the relationship between social media and remote work in a data-oriented and microcontent manner, whereas the second cluster addresses the economic effects of remote work in a macrooriented and theoretical manner. The final cluster examines the impact of AI and digital transformation on HRM from an organizational and strategic perspective. This three-part structure indicates both thematic diversity and temporal stratification within the field. New clusters are expected to emerge in the future to address emerging areas.

The findings of this bibliometric analysis carry significant theoretical and practical implications. Theoretically, this study maps the conceptual structure of an emerging interdisciplinary field, demonstrating how the convergence of technological and social disruptions (like the pandemic) creates new research paradigms. It extends existing theories on the future of work by providing empirical, large-scale evidence of how AI and flexibility are co-evolving in scholarly discourse. Practically, this research offers valuable insights for organizational leaders and HR professionals navigating the dual transformation of digitalization and workplace flexibility. The identified motor themes (remote work, AI, digitalization) serve as a strategic roadmap for managers prioritizing investments in technology and flexible work policies. Furthermore, the prominence of surveillance and work engagement as emerging themes highlights critical areas for organizational attention, balancing productivity with employee well-being in hybrid or remote settings.

In conclusion, this research provides a current and comprehensive overview of studies on artificial intelligence and flexible work arrangements, offering original and up-to-date information on general trends and the current state of the field. By synthesizing bibliographic findings and empirical contributions, this research contributes to the literature and provides insight and a foundation for future studies.

5. Final Considerations

This study offers a comprehensive and integrative overview of how “artificial intelligence” and “flexible work arrangements” have evolved and been addressed within the social sciences, particularly highlighting the surge of interest and scholarly output over the last decade. The findings underscore the growing significance of these concepts across diverse subfields, driven by transformative events such as the COVID-19 pandemic and advances in deep learning, and reveal their central role in shaping organizational structures, labor markets, and workforce adaptation in the digital era. While bibliometric trends demonstrate international engagement and thematic diversity, the relative novelty of the publications, dataset size, and reliance on the Web of Science database limit the immediate measurement of citation impact. Nevertheless, the study’s thematic analyses indicate that motor themes such as remote work, artificial intelligence, coronavirus, and digitalization will continue to define the research agenda, with emerging topics such as surveillance, work engagement, and machine learning likely to gain further relevance. Future research could benefit from expanding datasets to include multiple databases and from examining broader conceptual and interdisciplinary connections. Overall, this research contributes both an updated empirical basis and conceptual insight to the literature, strengthening the foundation for subsequent scholarly inquiry and fostering a deeper understanding of the intertwined trajectories of technology and flexible work in the social sciences.

This research has several limitations. The dataset is limited to documents in the Web of Science (WoS) database and only includes articles in the social sciences. Although the dataset required for the analysis of the research questions can be obtained from the WoS database, the Scopus database has also become quite reliable for creating datasets in recent years.

Expanding future studies to include databases such as Scopus and Google Scholar will be beneficial for increasing the scope of future research. In particular, similar studies conducted in the future will be useful in observing the development of the field.

Ethical Considerations

Not applicable.

Conflict of Interest

The authors declare that they have no conflicts of interest.

Funding

This research did not receive any financial support.

References

Adams-Prassl, A., Boneva, T., Golin, M., & Rauh, C. (2022). Work that can be done from home: Evidence on variation within and across occupations and industries. *Labour Economics, 74*, 1–13. https://doi.org/10.1016/j.labeco.2021.102083

Aleem, M., Sufyan, M., Ameer, I., & Mustak, M. (2023). Remote work and the COVID-19 pandemic: An artificial intelligence-based topic modeling and a future agenda. *Journal of Business Research, 154*, 1–16. https://doi.org/10.1016/j.jbusres.2022.113303

Ali, T., Hussain, I., Hassan, S., & Anwer, S. (2024). Examine how the rise of AI and automation affects job security, stress levels, and mental health in the workplace. *Bulletin of Business and Economics, 13*(2), 1180–1186.

Allen, T. D., Golden, T. D., & Shockley, K. M. (2015). How effective is telecommuting? Assessing the status of our scientific findings. *Psychological Science in the Public Interest, 16*(2), 40–68. https://doi.org/10.1177/1529100615593273

Almosawi, F., Aldoseri, N., & Al-Sartawi, A. (2024). The impact of artificial intelligence accounting systems on workforce productivity and job satisfaction. In J. Kacprzyk (Ed.), *Business analytical capabilities and artificial intelligence-enabled analytics: Applications and challenges in the digital era* (pp. 431–450). Springer Nature Switzerland.

Archambault, É., Campbell, D., Gingras, Y., & Larivière, V. (2009). Comparing bibliometric statistics obtained from the Web of Science and Scopus. *Journal of the American Society for Information Science and Technology, 60*(7), 1320–1326. https://doi.org/10.1002/asi.21062

Aria, M., & Cuccurullo, C. (2017). Bibliometrix: An R-tool for comprehensive science mapping analysis. *Journal of Informetrics, 11*(4), 959–975. https://doi.org/10.1016/j.joi.2017.08.007

Arruda, H., Silva, E. R., Lessa, M., Proença Jr., D., & Bartholo, R. (2022). VOSviewer and bibliometrix. *Journal of the Medical Library Association, 110*(3), 392–394.

Baldwin, R., & Okubo, T. (2023). Are software automation and teleworker substitutes? Preliminary evidence from Japan. *The World Economy, 47*(4), 1531–1556. https://doi.org/10.1111/twec.13496

Banh, L., & Strobel, G. (2023). Generative artificial intelligence. *Electronic Markets, 33*(1), 1–17.

Bar-Ilan, J. (2008). Informetrics at the beginning of the 21st century—A review. *Journal of Informetrics, 2*(1), 1–52. https://doi.org/10.1016/j.joi.2007.11.001

Bhardwaj, S., Chopra, R., & Pandita, D. (2025). Navigating the digital evolution of HRM: An integrative review. *International Journal of Organizational Analysis, 33*(11), 4305–4329. https://doi.org/10.1108/IJOA-10-2024-4895

Bornmann, L., & Leydesdorff, L. (2014). Scientometrics in a changing research landscape: Bibliometrics has become an integral part of research quality evaluation and has been changing the practice of research. *EMBO Reports, 15*(12), 1228–1232. https://doi.org/10.15252/embr.201439608

Bouamrane, M. M., & Mair, F. S. (2014). Implementation of an integrated preoperative care pathway and regional electronic clinical portal for preoperative assessment. *BMC Medical Informatics and Decision Making, 14*(1), Article 93. https://doi.org/10.1186/1472-6947-14-93

Boustani, N. M. (2022). Artificial intelligence impact on banks' clients and employees in an Asian developing country. *Journal of Asia Business Studies, 16*(2), 267–278. https://doi.org/10.1108/JABS-09-2020-0376

Broadus, R. N. (1987). Toward a definition of bibliometrics. *Scientometrics, 12*(5–6), 373–379.

Brooks, T. A. (1996). Review of *Dictionary of bibliometrics*, by V. Diodato. *The Library Quarterly, 66*(1), 103–105.

Callon, M., Courtial, J. P., & Laville, F. (1991). Co-word analysis as a tool for describing the network of interactions between basic and technological research: The case of polymer chemistry. *Scientometrics, 22*(1), 155–205. https://doi.org/10.1007/BF02019280

Callon, M., Courtial, J. P., Turner, W. A., & Bauin, S. (1983). From translations to problematic networks: An introduction to co-word analysis. *Social Science Information, 22*(2), 191–235. https://doi.org/10.1177/053901883022002003

Chelliah, J. (2017). Will artificial intelligence usurp white collar jobs? *Human Resource Management International Digest, 25*(3), 1–3. https://doi.org/10.1108/HRMID-11-2016-0152

Cheng, Z., & Jin, X. (2024). The impact of job anxiety caused by artificial intelligence substitution on organizational members' turnover intention: The mediating role of job insecurity and the moderating effect of job uncertainty. *Korean Public Administration Review, 38*(3), 67–103. https://doi.org/10.15843/kpapr.38.3.2024.6.67

Chiong, K., & Xie, Y. (2024). AI and job satisfaction: Evidence from Glassdoor employees' reviews. *SSRN Electronic Journal*. https://doi.org/10.2139/ssrn.4976340

Choudhary, S. (2016). A theoretical framework on flexible work schedules. *International Journal of Academic Research and Development, 1*(10), 8–14.

Choudhury, P., Foroughi, C., & Larson, B. (2021). Work-from-anywhere: The productivity effects of geographic flexibility. *Strategic Management Journal, 42*(4), 655–683. https://doi.org/10.1002/smj.3251

Chung, H., & van der Lippe, T. (2020). Flexible working, work–life balance, and gender equality: Introduction. *Social Indicators Research, 151*(2), 365–381. https://doi.org/10.1007/s11205-018-2025-x

Çivilidağ, A., & Durmaz, Ş. (2024). Examining the relationship between flexible working arrangements and employee performance: A mini review. *Frontiers in Psychology, 15*, Article 1398309. https://doi.org/10.3389/fpsyg.2024.1398309

Cobo, M. J., López-Herrera, A. G., Herrera-Viedma, E., & Herrera, F. (2011). An approach for detecting, quantifying, and visualizing the evolution of a research field: A practical application to the fuzzy sets theory field. *Journal of Informetrics, 5*(1), 146–166. https://doi.org/10.1016/j.joi.2010.10.002

Cooper, I. D. (2015). Bibliometrics basics. *Journal of the Medical Library Association, 103*(4), 217–218. https://doi.org/10.3163/1536-5050.103.4.013

Dabbous, A., Aoun Barakat, K., & Merhej Sayegh, M. (2022). Enabling organizational use of artificial intelligence: An employee perspective. *Journal of Asia Business Studies, 16*(2), 245–266. https://doi.org/10.1108/JABS-09-2020-0372

De Menezes, L. M., & Kelliher, C. (2017). Flexible working, individual performance, and employee attitudes: Comparing formal and informal arrangements. *Human Resource Management, 56*(6), 1051–1070. https://doi.org/10.1002/hrm.21822

Diodato, V. P., & Gellatly, P. (1994). *Dictionary of bibliometrics*. Routledge. https://doi.org/10.4324/9780203714133

Dwivedi, Y. K., Kshetri, N., Hughes, L., Slade, E. L., Jeyaraj, A., Kar, A. K., … Wright, R. (2023). Opinion paper: "So what if ChatGPT wrote it?" Multidisciplinary perspectives on opportunities, challenges and implications of generative conversational AI for research, practice and policy. *International Journal of Information Management, 71*, 1–63. https://doi.org/10.1016/j.ijinfomgt.2023.102642

Garfield, E. (2001). From the science of science to scientometrics: Visualizing the history of science with HistCite software. *Journal of Informetrics, 1*(2), 75–80. https://doi.org/10.1016/j.joi.2009.03.009

Garfield, E. (2004). Historiographic mapping of knowledge domains literature. *Proceedings of the National Academy of Sciences, 101*, 5183–5185. https://doi.org/10.1073/pnas.0307654100

Garfield, E., Pudovkin, A. I., & Istomin, V. S. (2003). Why do we need algorithmic historiography? *Journal of the American Society for Information Science and Technology, 54*(5), 400–412. https://doi.org/10.1002/asi.10226

Gursoy, D., & Cai, R. (2025). Artificial intelligence: An overview of research trends and future directions. *International Journal of Contemporary Hospitality Management, 37*(1), 1–17. https://doi.org/10.1108/IJCHM-03-2024-0322

Haenlein, M., & Kaplan, A. (2019). A brief history of artificial intelligence: On the past, present, and future of artificial intelligence. *California Management Review, 61*(4), 5–14. https://doi.org/10.1177/0008125619864925

Hayman, J. R. (2009). Flexible work arrangements: Exploring the linkages between perceived usability of flexible work schedules and work/life balance. *Community, Work & Family, 12*(3), 327–338. https://doi.org/10.1080/13668800902966331

Hou, Y., & Fan, L. (2024). Working with AI: The effect of job stress on hotel employees' work engagement. *Behavioral Sciences, 14*(11), Article 1076. https://doi.org/10.3390/bs14111076

Huang, M.-H., & Rust, R. T. (2018). Artificial intelligence in service. *Journal of Service Research, 21*(2), 155–172. https://doi.org/10.1177/1094670517752459

Ikpaahindi, L. (1985). An overview of bibliometrics: Its measurements, laws and their applications. *Libri, 35*(2), 163–177.

Janiesch, C., Zschech, P., & Heinrich, K. (2021). Machine learning and deep learning. *Electronic Markets, 31*(3), 685–695. https://doi.org/10.1007/s12525-021-00475-2

Kecklund, L. G., Beckers, D. G. J., Leineweber, C., & Tucker, P. T. (2017). How does work fit with my life? The relation between flexible work arrangements, work–life balance and recovery from work: An international perspective. In N. Chmiel, F. Fraccaroli, & M. Sverke (Eds.), *An introduction to work and organizational psychology* (pp. 430–447). John Wiley & Sons.

Kelley, P. G., Yang, Y., Heldreth, C., Moessner, C., Sedley, A., Kramm, A., … Woodruff, A. (2021). Exciting, useful, worrying, futuristic: Public perception of artificial intelligence in 8 countries. In *Proceedings of the 2021 AAAI/ACM Conference on AI, Ethics, and Society* (pp. 627–637). https://doi.org/10.1145/3461702.3462605

Kelliher, C., & Anderson, D. (2010). Doing more with less? Flexible working practices and the intensification of work. *Human Relations, 63*(1), 83–106. https://doi.org/10.1177/0018726709349199

Kliestik, T., Dragomir, R., Băluță, A. V., Grecu, I., Durana, P., Karabolevski, O. L., … Gajdosikova, D. (2024). Enterprise generative artificial intelligence technologies, Internet of Things and blockchain-based fintech management, and digital twin industrial metaverse in the cognitive algorithmic economy. *Oeconomia Copernicana, 15*(4). https://doi.org/10.24136/oc.3109

Kniffin, K. M., Narayanan, J., Anseel, F., Antonakis, J., Ashford, S. P., Bakker, A. B., … Van Vugt, M. (2021). COVID-19 and the workplace: Implications, issues, and insights for future research and action. *American Psychologist, 76*(1), 63–77. https://doi.org/10.1037/amp0000716

Kong, H., Yuan, Y., Baruch, Y., Bu, N., Jiang, X., & Wang, K. (2021). Influences of artificial intelligence (AI) awareness on career competency and job burnout. *International Journal of Contemporary Hospitality Management, 33*(2), 717–734. https://doi.org/10.1108/IJCHM-07-2020-0789

Kraus, S., Ferraris, A., & Bertello, A. (2023). The future of work: How innovation and digitalization re-shape the workplace. *Journal of Innovation & Knowledge, 8*(4), Article 100438. https://doi.org/10.1016/j.jik.2023.100438

Kurtovic, A., & Samardzija, J. (2023). Impact of artificial intelligence on employee mental health, productivity, and overall economic growth. In *96th International Scientific Conference on Economic and Social Development* (pp. 374–383).

Lakner, Z., Plasek, B., Kiss, A., Soós, S., & Temesi, Á. (2021). Derailment or turning point? The effect of the COVID-19 pandemic on sustainability-related thinking. *Sustainability, 13*(10), 5506. https://doi.org/10.3390/su13105506

Lawani, S. M. (1981). Bibliometrics: Its theoretical foundations, methods and applications. *Libri, 31*(1), 294–315. https://doi.org/10.1515/libr.1981.31.1.294

Lee, C. S., & Tajudeen, F. P. (2020). Usage and impact of artificial intelligence on accounting: Evidence from Malaysian organisations. *Asian Journal of Business and Accounting, 13*(1). https://doi.org/10.22452/ajba.vol13no1.8

Lestari, N. S., Rosman, D., & Millenia, E. (2023). The association between smart technology, artificial intelligence, robotics, and algorithms (STARA) awareness, job stress, job insecurity, and job satisfaction among hotel employees during the COVID-19 pandemic. In *E3S Web of Conferences* (Vol. 388, Article 03021). https://doi.org/10.1051/e3sconf/202338803021

Leuhery, F. (2024). The role of technology in employee training and development: A systematic review of recent advances and future directions. *Management Studies and Business Journal, 1*(3), 369–385. https://doi.org/10.62207/jmnzaw55

Lopez Soler, J. R., Christidis, P., & Vassallo, J. M. (2021). Teleworking and online shopping: Socio-economic factors affecting their impact on transport demand. *Sustainability, 13*(13), 7211. https://doi.org/10.3390/su13137211

Maguire, M. (2014). Socio-technical systems and interaction design: 21st century relevance. *Applied Ergonomics, 45*(2), 162–170. https://doi.org/10.1016/j.apergo.2013.05.011

McBurney, M. K., & Novak, P. L. (2002). What is bibliometrics and why should you care? In *Proceedings of the IEEE International Professional Communication Conference* (pp. 108–114). https://doi.org/10.1109/IPCC.2002.1049094

Mellner, C., Aronsson, G., & Kecklund, G. (2014). Boundary management preferences, boundary control, and work-life balance among full-time employed professionals in knowledge-intensive, flexible work. *Nordic Journal of Working Life Studies, 4*(4), 7–23. https://doi.org/10.19154/njwls.v4i4.4705

Nankervis, A., Connell, J., Cameron, R., Montague, A., & Prikshat, V. (2021). Are we there yet? Australian HR professionals and the Fourth Industrial Revolution. *Asia Pacific Journal of Human Resources, 59*(1), 3–19. https://doi.org/10.1111/1744-7941.12245

Pritchard, A., & Wittig, G. R. (1981). *Bibliometrics: A bibliography and index*. Allm Books.

Qamar, Y., Agrawal, R. K., Samad, T. A., & Chiappetta Jabbour, C. J. (2021). When technology meets people: The interplay of artificial intelligence and human resource management. *Journal of Enterprise Information Management, 34*(5), 1339–1370. https://doi.org/10.1108/JEIM-11-2020-0436

Riehmann, P., Hanfler, M., & Froehlich, B. (2005). Interactive Sankey diagrams. In *Proceedings of the IEEE Symposium on Information Visualization* (pp. 233–240). https://doi.org/10.1109/INFVIS.2005.1532152

Russell, H., O'Connell, P. J., & McGinnity, F. (2009). The impact of flexible working arrangements on work–life conflict and work pressure in Ireland. *Gender, Work & Organization, 16*(1), 73–97. https://doi.org/10.1111/j.1468-0432.2008.00431.x

Saura, J. R., Palacios-Marqués, D., Correia, M. B., & Barbosa, B. (2023). Innovative behavior in entrepreneurship: Analyzing new perspectives and challenges. *Frontiers in Psychology, 14*, Article 1123236. https://doi.org/10.3389/fpsyg.2023.1123236

Shagvaliyeva, S., & Yazdanifard, R. (2014). Impact of flexible working hours on work–life balance. *American Journal of Industrial and Business Management, 4*(1), 20–23. https://doi.org/10.4236/ajibm.2014.41004

Sharif, M. N., Zhang, L., Asif, M., Alshdaifat, S. M., & Hanaysha, J. R. (2025). Artificial intelligence and employee outcomes: Investigating the role of job insecurity and technostress in the hospitality industry. *Acta Psychologica, 253*, Article 104733. https://doi.org/10.1016/j.actpsy.2025.104733

Sharma, M., Biros, D., Baham, C., & Biros, J. (2024). What went wrong? Identifying risk factors for popular negative consequences in AI. *AIS Transactions on Human–Computer Interaction, 16*(2), 139–176. https://doi.org/10.17705/1thci.00203

Singh, A. K., Tyagi, P. K., Singh, A. K., Tyagi, P., Kapure, S., & Singh, E. R. (2022). Robotics and artificial intelligence in the hotel industry: A systematic literature review. In *Proceedings of the 8th International Conference on Advanced Computing and Communication Systems (ICACCS)* (pp. 1788–1792). https://doi.org/10.1109/ICACCS54159.2022.9785257

Singh, A., Jha, S., Srivastava, D. K., & Somarajan, A. (2022). Future of work: A systematic literature review and evolution of themes. *Foresight, 24*(1), 99–125. https://doi.org/10.1108/FS-09-2020-0093

Solanki, K. R. (2013). Flextime association with job satisfaction, work productivity, motivation, and employees' stress levels. *Journal of Human Resource Management, 1*(1), 9–14. https://doi.org/10.11648/j.jhrm.20130101.12

Subramaniam, A. G., Overton, B. J., & Maniam, C. B. (2015). Flexible working arrangements, work–life balance and women in Malaysia. *International Journal of Social Science and Humanity, 5*(1), 34–38. https://doi.org/10.7763/IJSSH.2015.V5.417

Subramony, M., Kepes, S., Yagil, D., Groth, M., & Solnet, D. (2023). The influence of technology on the future of work: Bibliometric review and directions for management scholarship. *Group & Organization Management*, Advance online publication, 1–29. https://doi.org/10.1177/10596011231212517

Thelwall, M. (2008). Bibliometrics to webometrics. *Journal of Information Science, 34*(4), 605–621. https://doi.org/10.1177/0165551507087238

Thompson, D. F., & Walker, C. K. (2015). A descriptive and historical review of bibliometrics with applications to medical sciences. *Pharmacotherapy: The Journal of Human Pharmacology and Drug Therapy, 35*(6), 551–559. https://doi.org/10.1002/phar.1586

Vaccaro, M., Almaatouq, A., & Malone, T. (2024). When combinations of humans and AI are useful: A systematic review and meta-analysis. *Nature Human Behaviour*, Advance online publication, 2293–2303. https://doi.org/10.1038/s41562-024-02024-1

Wijayati, D. T., Rahman, Z., Rahman, M. F. W., Arifah, I. D. C., & Kautsar, A. (2022). A study of artificial intelligence on employee performance and work engagement: The moderating role of change leadership. *International Journal of Manpower, 43*(2), 486–512. https://doi.org/10.1108/IJM-07-2021-0423

Yao, Q., Chen, K., Yao, L., Lyu, P., Yang, T., Luo, F., … Liu, Z. (2014). Scientometric trends and knowledge map of global health systems research. *Health Research Policy and Systems, 12*, Article 26. https://doi.org/10.1186/1478-4505-12-26

Zhang, C., Yu, M. C., & Marin, S. (2021). Exploring public sentiment on enforced remote work during COVID-19. *Journal of Applied Psychology, 106*(6), 797–810. https://doi.org/10.1037/apl0000933